

OE-EC804A — Artificial Intelligence

Semester VIII • 3L:0T:0P • 3 Credits

Module 3: Informed Search & Constraint Satisfaction Problems

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1. Informed (Heuristic) Search Strategies

Informed search strategies use domain-specific knowledge provided by a **Heuristic Function** $h(n)$ to guide the search towards goal states more efficiently.

Definition of a Heuristic Function

A **Heuristic Function** $h(n)$ estimates the cost of the cheapest path from state n to a goal state. For any goal state g , $h(g) = 0$.

1.1 Greedy Best-First Search

- **Evaluation Function:** $f(n) = h(n)$.
- Expands the node that appears to be closest to the goal.
- **Properties:** Incomplete in looped graphs; non-optimal. Time and Space complexity $O(b^m)$ worst-case.

1.2 A* Search Algorithm

A* combines the cost-so-far $g(n)$ and estimated remaining cost $h(n)$.

A* Evaluation Function

$$f(n) = g(n) + h(n) \quad (1)$$

where:

- $g(n)$: Exact path cost from initial state to node n .
- $h(n)$: Estimated cost from node n to goal state.
- $f(n)$: Estimated total cost of path through n to goal.

1.3 Conditions for A* Optimality

1. **Admissibility (Tree Search Optimality):** A heuristic $h(n)$ is admissible if it **never overestimates** the true cost to reach the goal:

$$h(n) \leq h^*(n) \quad \forall n \quad (2)$$

where $h^*(n)$ is the true minimum cost from n to goal.

2. **Monotonicity / Consistency (Graph Search Optimality):** A heuristic $h(n)$ is consistent if for every node n and successor n' generated by action a :

$$h(n) \leq c(n, a, n') + h(n') \quad (3)$$

This satisfies the triangle inequality and guarantees $f(n)$ is non-decreasing along any path.

1.4 On-line Search Agents and Unknown Environments

Unlike offline search algorithms (which compute a complete solution path before taking their first action), an **On-line Search Agent** interleaves computation and action: it acts, observes the new state, and decides its next move.

On-line Search Agent

An **On-line Search Agent** operates in unknown or dynamic environments by exploring state space physically. It computes an action, executes it in the environment, senses the resulting state s' , and updates its internal map.

- **Offline vs. On-line Search:** Offline agents compute the full path in a simulator before moving. On-line agents move in the physical world to discover state transitions and costs.
- **Competitive Ratio:** Evaluates on-line search performance by comparing actual path cost incurred to the optimal path cost if the environment were known in advance.
- **LRTA* (Learning Real-Time A*):** An on-line search algorithm that updates heuristic estimates $H(s)$ of visited states based on neighbor cost:

$$H(s) \leftarrow \min_{a \in \text{Actions}(s)} [c(s, a, s') + H(s')] \quad (4)$$

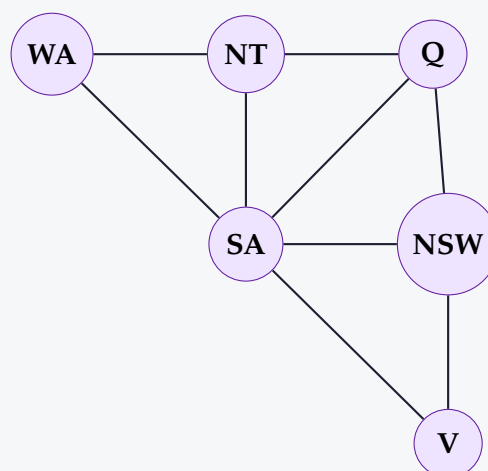
2. Constraint Satisfaction Problems (CSPs)

Definition of a CSP

A **Constraint Satisfaction Problem (CSP)** is defined by a triple (X, D, C) :

- $X = \{X_1, X_2, \dots, X_n\}$: Set of variables.
- $D = \{D_1, D_2, \dots, D_n\}$: Set of domains (possible values for each variable).
- $C = \{C_1, C_2, \dots, C_m\}$: Set of constraints specifying allowable value combinations.

Constraint Graph Representation (Map Coloring Example)



2.1 Backtracking Search for CSPs

Backtracking search is a depth-first search that chooses values for one variable at a time and backtracks when a variable has no legal values left.

Heuristics for CSP Backtracking

1. **Minimum Remaining Values (MRV):** Choose variable with fewest legal values remaining ("fail-first" heuristic).
2. **Degree Heuristic:** Choose variable involved in largest number of constraints on unassigned variables.
3. **Least Constraining Value (LCV):** Choose value that rules out fewest choices for neighboring variables.

2.2 Constraint Propagation & AC-3 Algorithm

Constraint propagation enforces local consistency to prune search spaces early.

- **Forward Checking:** When variable X is assigned, remove inconsistent values from domains of unassigned neighbors.
- **Arc Consistency (AC-3):** Arc $X_i \rightarrow X_j$ is consistent if for every value $x \in D_i$, there exists an allowed value $y \in D_j$.

2.3 Local Search for CSPs (Min-Conflicts Algorithm)

Unlike backtracking search which operates on partial assignments, local search algorithms for CSPs use a **complete-state formulation**: every variable is initially assigned a value (even if constraints are violated).

Min-Conflicts Heuristic

In each iteration, **Min-Conflicts** randomly selects a conflicted variable X_i and reassigns it the value $v \in D_i$ that minimizes the total number of constraint violations across the CSP.

- **Algorithm Mechanism:** Start with a complete assignment $\rightarrow$ pick a conflicted variable X at random $\rightarrow$ set X to value minimizing constraint violations $\rightarrow$ repeat until solution or step limit.
- **Performance:** Solves N -Queens for $N = 1,000,000$ in seconds; space complexity is $O(n)$ (linear).
- **Escaping Local Minima:** Uses random walks, simulated annealing, or random restarts to escape plateau states.

Quick Revision — Module 3

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Key Concepts and Metrics at a Glance

A* Evaluation Function	$f(n) = g(n) + h(n)$ ($g(n)$ cost-so-far, $h(n)$ estimated remaining cost)
Admissibility Condition	$h(n) \leq h^*(n)$ (Heuristic never overestimates true cost to goal)
Consistency Condition	$h(n) \leq c(n, a, n') + h(n')$ (Satisfies triangle inequality along edges)
CSP Formalization	Triple (X, D, C) of Variables, Domains, and Constraints
Min-Conflicts Heuristic	Local search reassigning conflicted variable value to minimize violations.

One-Line Definitions

- **Admissible Heuristic:** A heuristic that never overestimates the actual cost to reach goal.
- **Consistent Heuristic:** A heuristic obeying triangle inequality along graph edges.
- **Arc Consistency (AC-3):** State where every value in a variable domain has a valid partner in neighbor domains.
- **MRV Heuristic:** Chooses the variable with the fewest remaining legal domain choices.

Frequently Asked / Viva Questions

Q1. Define A* search. What is its evaluation function?

A* evaluates nodes using $f(n) = g(n) + h(n)$, combining cost-so-far $g(n)$ and heuristic estimate $h(n)$.

Q2. What is an Admissible Heuristic?

A heuristic $h(n)$ that never overestimates true cost to goal ($h(n) \leq h^*(n)$).

Q3. Differentiate between Admissibility and Consistency.

Admissibility guarantees optimality for tree search. Consistency ($h(n) \leq c(n, a, n') + h(n')$) guarantees $f(n)$ is non-decreasing, ensuring optimality for graph search without re-opening nodes.

Q4. Define CSP formally.

A triple (X, D, C) where X is set of variables, D set of domains, and C set of constraints.

Q5. Explain the MRV heuristic in CSPs.

Frequently Asked / Viva Questions (contd.)

Minimum Remaining Values heuristic chooses the variable with fewest valid remaining choices, aiming to fail early.

Q6. What is Forward Checking?

In CSP search, whenever variable X is assigned, any inconsistent values are pruned from neighboring unassigned variables.

Q7. What is the AC-3 algorithm?

An algorithm that enforces arc consistency across all variable pairs in a CSP constraint graph.

Q8. Compare Greedy Best-First Search and A* Search.

Greedy uses $f(n) = h(n)$ (fast, non-optimal). A* uses $f(n) = g(n) + h(n)$ (optimal if h is admissible).

Q9. What is Manhattan Distance heuristic in 8-puzzle?

Sum of absolute differences in horizontal and vertical coordinates of tiles from goal positions.

Q10. Why is LCV used for value selection in CSPs?

Least Constraining Value maximizes flexibility for remaining variables, leaving maximal choices open.